Exercise Set 3

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In the lecture, you have seen Coppersmith's method for finding small roots of $F(x) \equiv 0 \pmod{M}$: polynomials are turned into vectors, LLL finds a short vector, and Howgrave-Graham's theorem turns it back into a polynomial with the root over the integers. Problem 1 makes the correspondence between polynomials and vectors precise. Problem 2 extends the method to roots modulo an unknown divisor of N, i.e. the setting of the motivating example of the lectures where the modulus is the unknown p and only $N = pq$ is available.

Problem 1 (Polynomial-vector transformations). Denote by $\mathbb{Z}[x]$ the ring of polynomials with integer coefficients and fix a positive integer X. In the lectures, we have defined a way to convert between polynomials of degree at most d in $\mathbb{Z}[x]$ and (row) vectors in $\mathbb{Z}^{d+1}$.

Consider the following function:

$$\begin{aligned} \mathsf{poly2vec}: \quad & \mathbb{Z}[x] \to \mathbb{Z}^{d+1} \\ & a_0 + a_1 x + a_2 x^2 + \ldots + a_d x^d \mapsto [a_0, a_1 X, a_2 X^2, \ldots, a_d X^d] \end{aligned}$$

Let $\mathsf{vec2poly}$ be the following operation:

$$\begin{aligned} \mathsf{vec2poly}: \quad & \mathbb{Z}^{d+1} \to \mathbb{Z}[x] \\ & [v_0, v_1, v_2, \ldots, v_d] \mapsto v_0 + \frac{v_1}{X} x + \frac{v_2}{X^2} x^2 + \ldots + \frac{v_d}{X^d} x^d \end{aligned}$$

(a) Consider a polynomial $F(x) \in \mathbb{Z}[x]$. Prove that the result of $\mathsf{poly2vec}(F)$ can be computed by taking the coefficient vector of $F(xX)$ (i.e. the polynomial F evaluated at $x \cdot X$).

Solution. Let $F(x) = a_0 + a_1 x + \cdots + a_d x^d$. Then $F(Xx) = a_0 + a_1 Xx + a_2 (Xx)^2 + \ldots + a_d (Xx)^d = a_0 + a_1 Xx + a_2 X^2 x^2 + \ldots + a_d X^d x^d$, whose coefficient vector is $[a_0, a_1 X, \ldots, a_d X^d] = \mathsf{poly2vec}(F)$.

(b) Argue that the function $\mathsf{vec2poly}$ is well defined. That is, prove that, for any polynomial $F(x)$ of degree at most d, $\mathsf{vec2poly}(\mathsf{poly2vec}(F)) = F$ and that all operations are well defined.

Solution. Since we are in a ring, not all divisions are necessarily well defined. We first have to prove that we can always divide by X. This is easy to see, as the result of $\mathsf{poly2vec}$ is $[v_0, v_1, \ldots, v_d] = [a_0, a_1 X, \ldots, a_d X^d]$, which means that v_i is always divisible by X^i. Proving that $\mathsf{vec2poly}$ is the inverse operation of $\mathsf{poly2vec}$ can be done by noting that the latter multiplies the coefficient of x^i by X^i and collects the results in a vector, while $\mathsf{vec2poly}$ divides the i-th entry by X^i again, i.e. does exactly the opposite operation.

(c) Consider two polynomials $F_1(x), F_2(x) \in \mathbb{Z}[x]$ of degree at most d with a common root x_0 and their corresponding vectors $\mathbf{v}_i := \mathsf{poly2vec}(F_i)$, for $i \in \{1, 2\}$. Prove that the following operations are well-defined and their results are polynomials that also have a root at x_0.

Solution. We start by defining common notation. Let $F_1 = \sum_{i=0}^{d} a_i x^i$ and $F_2 = \sum_{i=0}^{d} b_i x^i$. We assume that if either polynomial has degree $d' < d$, all coefficients with index in $\{d' + 1, ..., d\}$ are 0. Then:

$$\mathsf{poly2vec}(F_1) = [a_0, a_1 X, \ldots, a_d X^d]$$

$$\mathsf{poly2vec}(F_2) = [b_0, b_1X, \ldots, b_dX^d]$$

- $\mathsf{vec2poly}(\mathsf{poly2vec}(F_1) + \mathsf{poly2vec}(F_2))$

  Solution.

  $$\mathsf{poly2vec}(F_1) + \mathsf{poly2vec}(F_2) = [a_0 + b_0, (a_1 + b_1)X, \ldots, (a_d + b_d)X^d]$$

  The latter corresponds to a new polynomial $F_{\text{sum}}(x) = (a_0 + b_0) + (a_1 + b_1)x + \ldots + (a_d + b_d)x^d = (F_1 + F_2)(x)$. Thus, this transformation simply sums two polynomials. We know that if F_1 and F_2 share a root x_0, then their sum will also have that root: $F_1(x_0) = 0 \wedge F_2(x_0) = 0 \Rightarrow F_1(x_0) + F_2(x_0) = (F_1 + F_2)(x_0) = 0$.

- $\mathsf{vec2poly}(k \cdot \mathsf{poly2vec}(F_1))$ for $k \in \mathbb{Z}$

  Solution.

  $$k \cdot \mathsf{poly2vec}(F_1) = [k \cdot a_0, k \cdot a_1X, \ldots, k \cdot a_dX^d]$$

  $$\mathsf{vec2poly}(k \cdot \mathsf{poly2vec}(F_1)) = ka_0 + ka_1x + \ldots + ka_dx^d = k \cdot F_1(x)$$

  If $F_1(x_0) = 0$, then $k \cdot F_1(x_0) = 0$.

(d) Consider $\mathbb{Z}_N$, where N is an RSA modulus. Assume we use Coppersmith's method to find a small root x_0 of a polynomial $F(x) \in \mathbb{Z}_N[x]$. Argue that the matrix *after* the application of LLL consists of rows, all of which encode polynomials that have x_0 as a root modulo N.

Solution. The first step is to visualize $F(x)$ as a polynomial in $\mathbb{Z}[x]$ rather than $\mathbb{Z}_N[x]$. Recall that all elements of $\mathbb{Z}_N$ are equivalence classes modulo N. Each equivalence class is the set of all numbers that are equivalent to each other modulo N, e.g. $[3] = \{3, 3+N, 3-N, 3+2N, ...\} \subset \mathbb{Z}$. The natural way of projecting the polynomial to $\mathbb{Z}[x]$ is to map each coefficient $[k] \in \mathbb{Z}_N$ to its smallest non-negative representative $k \in \mathbb{Z}$.

Next, recall that Coppersmith's method begins by constructing a set of polynomials all having x_0 as a root modulo N, then converting each of the polynomials to a vector. These vectors form the rows of the basis matrix B of a lattice $\mathcal{L}$. Then, when LLL is applied to B, it will only apply elementary row operations. By the previous point, these operations yield vectors that, when converted back into polynomials in $\mathbb{Z}[x]$, preserve the root x_0 modulo N. Only for a sufficiently short vector does Howgrave-Graham's theorem turn this into a root over the integers.

Problem 2 (Generalizing Coppersmith's Method). In the lecture you have seen that Coppersmith's method can be used to find small roots of a polynomial $F(x)$ modulo N, even when the factorization of N is not known. In this exercise, we show that Coppersmith's method can also be employed to find small roots of a polynomial $F(x)$ *modulo a divisor of N, even if we do not know the divisor*. This, however, requires changing the algorithm a bit.

Let N be a large integer of unknown factorization.[1] Let b be an unknown divisor of N with

[1] Recall that, if the factorization is known, one can split the problem into sub-problems, one for each factor. Each sub-problem consists of finding a root modulo a prime p, which is easy. One can then compose the final solution by using the Chinese Remainder Theorem.

$b \geq N^\beta$ for some known $0 < \beta \leq 1$.[2] Let $F(x) \in \mathbb{Z}[x]$ be a monic polynomial of degree d with $F(x_0) \equiv 0 \pmod{b}$ for some x_0 with $|x_0| < X$, $X \in \mathbb{Z}$. Let $h \geq 2$ be an integer to be fixed later.

Consider the sequence of polynomials from the "Full Coppersmith Method" from the lectures:

$$G_{i,j}(x) = N^{h-1-j} F(x)^j x^i, \quad 0 \leq i < d,\ 0 \leq j < h$$

Note that all $G_{i,j}(x_0) \equiv 0 \pmod{b^{h-1}}$ and that $G_{i,j}$ has degree $dj + i$. Let $\mathcal{L}$ be the lattice with $\mathsf{poly2vec}(G_{i,j})$ as basis vectors. Note that $\mathcal{L}$ has dimension dh.[3]

(a) Consider $F(x) = x + a$. Assume that $F(x_0) \equiv 0 \pmod{b}$, where $b \geq N^{\frac{1}{2}}$ (thus $\beta = 1/2$) and $|x_0| < X$ for some $X \in \mathbb{Z}$. Show that the method above cannot provably find this small root x_0, even for large values of h (e.g. $h = 50$).

Hint: this requires you to show that there does not exist any value of X that satisfies the Howgrave-Graham condition.

Solution. We build the matrix as above. Note that the degree of F is $d = 1$. The polynomials will all be of the form:

$$G_j(x) = N^{h-1-j}(x + a)^j, \quad 0 \leq j < h$$

which induce the following basis matrix (we omit all the terms below the diagonal):

$$B = \begin{bmatrix} N^{h-1} & 0 & 0 & 0 & \dots & 0 & 0 \\ - & N^{h-2}X & 0 & 0 & \dots & 0 & 0 \\ - & - & N^{h-3}X^2 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ - & - & - & - & - & NX^{h-2} & 0 \\ - & - & - & - & - & - & X^{h-1} \end{bmatrix}$$

The corresponding lattice $\mathcal{L}$ (of dimension $n = h$) will have a determinant $\det(\mathcal{L}) = (NX)^{(h-1)+\dots+1} = (NX)^{h(h-1)/2}$. By the LLL guarantees, we get that, after basis reduction, the first vector of the basis $\mathbf{b_1}$ will have a length of at most:

$$\|\mathbf{b_1}\| \leq 2^{(n-1)/4} \det(\mathcal{L})^{1/n} = 2^{(h-1)/4} (NX)^{h(h-1)/(2h)} = 2^{(h-1)/4} (NX)^{(h-1)/2}$$

In order to have a guaranteed solution, we need this vector to respect Howgrave-Graham's condition. Because all the $G_{i,j}(x_0) \equiv 0 \pmod{b^{h-1}}$, then any polynomial corresponding to a vector in the reduced basis will also vanish at x_0 modulo b^{h-1}, so we need $\|\mathbf{b_1}\| < b^{h-1}/\sqrt{dh}$. Since b is unknown, the only thing we can use is $b \geq N^\beta$, so the condition we can actually guarantee is

$$2^{(h-1)/4} (NX)^{(h-1)/2} < \frac{(N^\beta)^{h-1}}{\sqrt{h}} = \frac{N^{\frac{h-1}{2}}}{\sqrt{h}} \leq \frac{b^{h-1}}{\sqrt{dh}}$$

which, in turn, requires that:

[2] Note that the case $\beta = 1$ corresponds to the case already analyzed in the lectures.

[3] The maximum degree of any polynomial is $d \cdot (h-1) + (d-1) = dh - 1$, which means that there are $dh - 1 + 1 = dh$ coefficients

$$X < \frac{1}{\sqrt{2} \cdot h^{1/(h-1)}} < 1$$

since $h^{-1/(h-1)} \leq 1$ for every $h \geq 2$. Since X is a positive integer, the upper bound is too small for every choice of h and does not allow for any value of X.

(b) Let $t = \lfloor dh(1/\beta - 1) \rfloor$. Consider now the following additional polynomials:

$$H_i(x) = x^i F(x)^h, \quad 0 \leq i < t$$

Once again, note how $H_i(x_0) \equiv 0 \pmod{b^{h-1}}$. Show that for $h = 2$, including both polynomials $G_{i,j}$ and H_i, we can provably find the root x_0 for the polynomial above if $X < N^{\frac{1}{6}}/2^{\frac{7}{6}}$.

Solution. By plugging in $d = 1$, $h = 2$ and $\beta = 1/2$, we get that $t = 2$. Note that the i-th polynomial H_i has degree $dh + i$. Adding these two polynomials gives us the following matrix:

$$B = \left[\begin{array}{cccc} N & 0 & 0 & 0 \\ - & X & 0 & 0 \\ \hline - & - & X^2 & 0 \\ - & - & - & X^3 \end{array}\right]$$

The corresponding lattice $\mathcal{L}$ has dimension $n = dh + t = 4$ and determinant $\det(\mathcal{L}) = N \cdot X^6$.

Once again, by the LLL guarantees, we get that, after basis reduction, the first vector of the basis $\mathbf{b_1}$ will have a length of at most:

$$\|\mathbf{b_1}\| \leq 2^{(n-1)/4} \det(\mathcal{L})^{1/n} = 2^{3/4} N^{1/4} X^{6/4}$$

We want this vector to respect Howgrave-Graham's condition with modulus b^{h-1}. Using $b \geq N^\beta$, it suffices that

$$2^{3/4} N^{1/4} X^{6/4} < \frac{(N^\beta)^{h-1}}{\sqrt{dh+t}} = \frac{N^{\frac{1}{2}}}{\sqrt{4}} = \frac{N^{\frac{1}{2}}}{2} \leq \frac{b^{h-1}}{\sqrt{dh+t}}$$

which, in turn, implies that:

$$X < \frac{N^{\frac{1}{6}}}{2^{\frac{7}{6}}}$$

Concretely, for a 1024-bit modulus N, this allows us to recover roots of up to about 169 bits.

Abstracting away from the concrete numbers, there are two general methods for improving the performance of Coppersmith's method (i.e. increasing the possible value of X).

The first one is to increase the dimension of the lattice, which is done by adding more polynomials like the H_i above. These are called x-shift polynomials, and they increase the dimension of the lattice without significantly increasing the determinant. The second method is to increase the modulus in the Howgrave-Graham condition. This is done by using larger powers of $F(x)$, since if $F(x_0) \equiv 0 \pmod{b}$, then $F(x_0)^k \equiv 0 \pmod{b^k}$. This is what the original polynomials $G_{i,j}$ do.